BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: KT-IX-22-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** Code B

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and overwriting will not be awarded any marks.

Qi. The diagonal of ..... does not divide it into two congruent triangles.

(A) rectangle (B) square

(C) parallelogram (D) trapezium

Qii. A diagonal of a parallelogram divides the parallelogram into .....congruent triangles.

(A) one (B) two

(C) three (D) None of these

Qiii. Perpendicular bisectors of a triangle are .....

(A) congruent (B) concurrent

(C) parallel to each other (D) perpendicular

Qiv. Determinant of [4 -1; -9 2] is ....

(A) –1 (B) 1

(C) 17 (D) –17

Qv. Factors of $x^2 + 2x - 24$ are .....

(A) x + 4, x – 6 (B) x – 4, x + 6

(C) x + 3, x – 8 (D) x + 8, x – 3

Qvi. The value of polynomial $P(x) = x^3 - x^2 - 5x + 3$ at x = 3 is

(A) –15 (B) 27

(C) 0 (D) 9

Qvii. Simplified form of $a/(25a^2–b^2) – b/(5a–b)$ is……

(A) $5a/(25a^2–b^2)$ (B) $+5a/(5a–b)$

(C) $–5a/(5a+b)$ (D) $–5a/(25a^2–b^2)$

Qviii. If x, y and z are any three real numbers then, if x < y and y < z => x < z is called …..property.

(A) trichotomy (B) additive

(C) transitive (D) multiplicative

Qix. The two coordinate axes intersect at an angle of ….

(A) 30° (B) 60°

(C) 45° (D) 90°

Qx. The points (-3,4) lies in which quadrant?.

(A) II (B) I

(C) IV (D) III

Qxi. A matrix with "m" rows and "n" columns has order ……

(A) m/n (B) m × n

(C) m^n (D) m – n

Qxii. √3 is an example of ……. number.

(A) rational (B) irrational

(C) prime (D) natural

Qxiii. $\sqrt{(-(-9)^2)}$= ?

(A) 9 (B) 9 + i

(C) 9–i (D) 9i

Qxiv. 0.069 can be written in scientific notation as.

(A) 6.9×10^2 (B) 6.9×10^{-3}

(C) 0.69×10^3 (D) 69×10^2

Qxv. $(a+b)^2 – (a–b)^2$ = ……

(A) $a^4–b^4$ (B) $2(a^2+b^2)$

(C) $a^2–4ab+2b^2$ (D) 4ab

Section "B"

Maximum Marks: 36

QQ.2. Attempt any NINE parts of the following. All parts carry equal marks.

i Find the multiplicative inverse of Z = [3 4; 1 2]

ii Simplify (x^a/x^b)^(a+b) · (x^b/x^c)^(b+c) · (x^c/x^a)^(c+a)

iii Find the volume of "a" log_√5 3a = log_√5 9 + log_√5 2 - log_√5 3

iv Solve the following system of linear equation using inversion method. X – 2y – 1 = 0, 2x + y + 3 = 0

v Divide $(x^2+x-2)/(3x^2+9x+6) \div (x - 1)$

vi If $x = \sqrt{3} - \sqrt{2}$, find the values of $x - 1/x$ and $x^2 + 1/x^2$

vii Factorize $x^4 + 1/x^4 - 7$

viii Find L.C.M of $x^5 - x$, $x^5 - x^2$ and $x^5 - x^3$

ix Find the square root of $(x^2 - 1/x^2)^2 - 12(x^2 - 1/x^2) + 36$

x Ali thought of a number, added 3 with it. Then he doubled the sum, he got 40. What was the original number?

xi Solve for x, | -5x + 1 | = 6

xii Draw the graph of y = x

Section "C"

Maximum Marks: 24

Note: Note: Attempt any THREE questions of the following. All questions carry equal marks.

QQ.3. Prove that if three are more parallel lines make congruent intercepts on a transversal, they also intercept congruent segments on any other line that cuts them.

QQ.4. Prove that any point on the right bisector of a line segment is equidistant from end points of the segment.

QQ.5. Prove that the line segment joining the mid points of two sides of a triangle is equal to half of the length of the third side.

QQ.6. Construct ΔKLM such that mKL = 6cm, m∠K = 70° and m∠M = 50°. Draw its medians and verify their concurrency.